

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ASSIGNMENT / ASSESSMENT

CSA0709 – COMPUTER NETWORKS

Design and Implementation of an FTP-Based File-Sharing Network Using Cisco Packet Tracer

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Section / Slot	D
Course Code & Name	CSA0709 – COMPUTER NETWORKS
Assignment Title	Design an FTP-based file-sharing network using Packet Tracer. Configure the FTP service and test communication between clients and server.

1. Introduction

File sharing is an important requirement in an organizational computer network. An FTP (File Transfer Protocol) server provides a centralized location where authorized clients can upload, download, and manage files. This assessment designs and implements a simple FTP-based file-sharing network using Cisco Packet Tracer. The network contains a server and multiple client systems connected through a switch. IP addressing, FTP service configuration, user access, connectivity testing, and file-transfer operations are performed and verified.

2. Problem Statement

Design and implement an FTP-based file-sharing network using Cisco Packet Tracer for an organization that requires centralized file sharing between multiple client systems and a server. Configure the network topology, assign appropriate IP addresses, configure the FTP service on the server, and create suitable user accounts and access permissions. Connect multiple clients to the FTP server and test FTP login, file upload, file download, and communication. Verify the network using appropriate connectivity tests and ensure that the designed system provides reliable and organized file-sharing services.

3. Objectives

- To design a client-server network using Cisco Packet Tracer.
- To configure suitable IPv4 addresses for the server and client systems.
- To configure the FTP service on the server.
- To create authorized FTP user accounts and permissions.
- To test connectivity between clients and the FTP server.
- To verify FTP login, file upload, and file download operations.

- To analyze the benefits of centralized file sharing and relate the activity to SDG 9.

4. Course Outcomes Addressed

CO	Outcome
CO1	Identify and explain principles of application-layer protocols and client-server communication.
CO2	Configure IP addressing, network devices, and communication parameters required to establish a client-server network using Cisco Packet Tracer.
CO3	Analyze FTP communication, client-server connectivity, and file-transfer operations using appropriate network simulation and packet-analysis tools.
CO4	Analyze FTP communication, client-server connectivity, and file-transfer operations using appropriate network simulation and packet-analysis tools.
CO5	Design and implement an FTP-based file-sharing network by configuring an FTP server, connecting multiple clients, performing file-transfer operations, and verifying communication using Cisco Packet Tracer.

5. Bloom's Taxonomy

The assessment addresses L6 – Design, L5 – Evaluate, L4 – Analyze, L3 – Configure, and L2 – Explain.

6. SDG Mapping

The assessment is mapped to SDG 4 – Quality Education and SDG 9 – Industry, Innovation and Infrastructure. The practical Packet Tracer activity develops networking and digital-infrastructure skills. In particular, the centralized FTP model demonstrates how organized network infrastructure can improve resource accessibility and support efficient information sharing.

7. Proposed Network Topology

A star topology is used. One switch acts as the central connecting device. The FTP server and three client PCs are connected to the switch using Ethernet connections. The server provides centralized file-sharing services to all authorized clients.

Topology:

- FTP Server (192.168.10.10) — Switch
- PC0 (192.168.10.11) ————— Switch

- PC1 (192.168.10.12) ————— Switch
- PC2 (192.168.10.13) ————— Switch

8. Devices Used

S.No	Device	Quantity	Purpose
1	Cisco 2960 Switch	1	Central network connection
2	Server-PT	1	FTP server
3	PC-PT	3	FTP clients
4	Copper Straight-Through Cable	4	Ethernet connections

9. IP Addressing Plan

Device	IPv4 Address	Subnet Mask	Default Gateway
FTP Server	192.168.10.10	255.255.255.0	192.168.10.1
PC0	192.168.10.11	255.255.255.0	192.168.10.1
PC1	192.168.10.12	255.255.255.0	192.168.10.1
PC2	192.168.10.13	255.255.255.0	192.168.10.1

Because all devices are in the same 192.168.10.0/24 LAN, the clients can communicate directly with the server through the switch. The gateway is included as a standard network configuration; communication within the same subnet does not require routing.

10. Cisco Packet Tracer Configuration Procedure

Step 1 – Build the topology: Place one 2960 switch, one Server-PT, and three PC-PT devices. Connect each end device to the switch using Copper Straight-Through cables.

Step 2 – Configure the server: Open Server-PT → Desktop → IP Configuration. Set IP address to 192.168.10.10, subnet mask to 255.255.255.0, and default gateway to 192.168.10.1.

Step 3 – Configure clients: Configure PC0 as 192.168.10.11, PC1 as 192.168.10.12, and PC2 as 192.168.10.13. Use subnet mask 255.255.255.0 and gateway 192.168.10.1.

Step 4 – Enable FTP: Open Server-PT → Services → FTP and switch the FTP service ON.

Step 5 – Create users: Add authorized accounts such as username 'student1' with password 'cisco123', and username 'student2' with password 'cisco123'. Enable the permissions required for file transfer, including read and write where appropriate.

Step 6 – Test connectivity: From each client, open Command Prompt and use ping 192.168.10.10. Successful replies confirm IP connectivity to the FTP server.

Step 7 – Test FTP login: From a client Command Prompt, enter ftp 192.168.10.10. Provide the configured username and password when prompted.

Step 8 – Test file upload: After successful FTP login, use an FTP PUT operation with a test file to verify that the client can upload a file to the server.

Step 9 – Test file download: Use an FTP GET operation with an available server file to verify that the client can download a file.

11. FTP Commands Used for Testing

Command	Purpose
ping 192.168.10.10	Checks IP connectivity from a client to the FTP server.
ftp 192.168.10.10	Starts an FTP session with the server.
dir	Displays files available through the FTP session.
put test.txt	Uploads test.txt from the client to the FTP server.
get test.txt	Downloads test.txt from the FTP server to the client.
bye	Closes the FTP session.

12. Testing and Expected Results

Test	Procedure	Expected Result
Client → Server ping	PC0/PC1/PC2 ping 192.168.10.10	Successful replies / connectivity
FTP login	ftp 192.168.10.10 + valid credentials	Login successful
Directory listing	dir	Server files displayed
File upload	put test.txt	File transferred to server
File download	get test.txt	File transferred to client

13. Analysis of FTP Communication

FTP is an application-layer protocol used for transferring files between a client and server. In this design, the server provides the centralized file-sharing service while the PCs act as clients. A client first establishes connectivity to the server, authenticates using an FTP account, and then performs file operations according to its assigned permissions. The Packet Tracer simulation makes it possible to observe and verify the logical communication between the client systems and the FTP server.

14. Digital Infrastructure and Organizational Benefits

- Centralized storage makes shared resources easier to locate and access.
- Authorized user accounts provide controlled access to files.
- A dedicated server simplifies administration of shared data.
- Multiple clients can use the same file-sharing service over the network.
- Centralized services can reduce duplication of files across individual client computers.

- The structured network provides a foundation for reliable organizational communication and resource sharing.

15. SDG Relevance – SDG 9

The activity supports SDG 9 – Industry, Innovation and Infrastructure by providing practical experience in designing and configuring digital network infrastructure. The FTP server, switch, addressing scheme, and client systems form a basic digital infrastructure for centralized information sharing. Learning to configure, test, and troubleshoot such systems develops skills relevant to reliable and organized ICT infrastructure.

16. Learning and Skills Demonstrated

- Understanding client-server architecture.
- IPv4 addressing and subnet configuration.
- Basic LAN and switch-based network design.
- FTP service configuration.
- User authentication and access permissions.
- Connectivity testing using ping.
- FTP file upload and download testing.
- Basic troubleshooting and verification.

17. Design Decisions and Justification

A star topology was selected because it provides a simple and organized connection between several clients and a central network device. A dedicated FTP server was selected to provide centralized file-sharing services. A /24 subnet was used so that the server and clients can communicate easily within the same LAN. Static IP addressing makes the Packet Tracer configuration easy to understand and simplifies troubleshooting. Separate FTP user accounts allow access to be controlled according to organizational requirements.

18. Troubleshooting

- If ping fails, verify cable connections, interface status, IP address, and subnet mask.
- If FTP login fails, verify that the FTP service is ON and that the username/password are correct.
- If file upload or download fails, verify the user's FTP permissions.
- If a client cannot communicate with the server, check that both devices are configured in the same subnet.
- Use Packet Tracer simulation mode when necessary to analyze packet flow and identify communication problems.

19. Reflection

This assessment demonstrates how a centralized client-server design can be implemented using Cisco Packet Tracer. The topology and IP addressing were selected to keep the network simple, organized, and easy to troubleshoot. Configuring the FTP service and user accounts showed how application-layer services operate over a network. Connectivity and file-transfer tests provided practical verification of the design. The activity also showed the importance of reliable digital infrastructure and its relationship to SDG 9. Overall, the assessment strengthened skills in network configuration, FTP operation, testing, troubleshooting, and technical communication.

20. Conclusion

An FTP-based file-sharing network was designed using a switch, one FTP server, and multiple client systems. The devices were assigned IP addresses, the FTP service was configured, and user accounts were created for controlled access. Ping and FTP tests were used to verify communication, login, file upload, and file download. The completed design demonstrates the role of centralized file sharing in an organizational network and provides practical experience in digital infrastructure configuration.

21. Submission Checklist

- ☐ Network topology created in Cisco Packet Tracer
- ☐ IP addresses configured
- ☐ FTP service enabled
- ☐ FTP user accounts created
- ☐ Client-server ping tested
- ☐ FTP login tested
- ☐ File upload tested
- ☐ File download tested
- ☐ Screenshots added from Packet Tracer
- ☐ SDG 9 relevance explained
- ☐ Reflection and conclusion completed